Topic: Adding power series

Question: Are the series in phase?

$$\sum_{n=1}^{\infty} nc_n x^{n-1}$$

$$3\sum_{n=0}^{\infty} c_n x^{n+2}$$

Answer choices:

A Yes, because the series begin at the same power of x

B Yes, because the series have matching indices

C No, because the series don't begin at the same power of x

D No, because the series don't have matching indices

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Solution: C

Two series are in phase when they begin at the same power of x. The index of the first series

$$\sum_{n=1}^{\infty} nc_n x^{n-1}$$

is $n = 1$, which means the first term of the series will be

$$1c_1 x^{1-1}$$

$$c_1 x^0$$

The index of the second series

$$3\sum_{n=0}^{\infty} c_n x^{n+2}$$

is $n = 0$, which means the first term of the series will be

$$3c_0 x^{0+2}$$

$$3c_0 x^2$$

The first term of the first series is for x^0, while the first term of the second series is for x^2. Because each series begins at a different power of x, the series are not in phase.

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Question: Find the difference of the power series.

$$\sum_{n=1}^{\infty} nc_n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n$$

Answer choices:

A $$\sum_{k=0}^{\infty} ((k+1)c_{k+1} + c_k)x^k$$

B $$\sum_{k=0}^{\infty} ((k+1)c_{k+1} - c_k)x^k$$

C $$\sum_{k=0}^{\infty} (kc_k + c_{k+1})x^k$$

D $$\sum_{k=0}^{\infty} (kc_k - c_{k+1})x^k$$

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Solution: B

Both series begin with the x^0 term, so they're already in phase. To make their indices match, we'll make the substitution $k = n - 1$ and $n = k + 1$ into the first series, and make the substitution $k = n$ into the second series.

$$\sum_{n=1}^{\infty} nc_n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n$$

$$\sum_{k=0}^{\infty} (k+1)c_{k+1} x^k - \sum_{k=0}^{\infty} c_k x^k$$

The series are now in phase with matching indices, so we'll add them and then factor out x^k.

$$\sum_{k=0}^{\infty} (k+1)c_{k+1} x^k - c_k x^k$$

$$\sum_{k=0}^{\infty} ((k+1)c_{k+1} - c_k)x^k$$

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Topic: Adding power series

Question: Find the sum of the power series.

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} - 2\sum_{n=1}^{\infty} nc_n x^n + \sum_{n=0}^{\infty} c_n x^n$$

Answer choices:

A $$\sum_{k=1}^{\infty} ((k+2)(k+1)c_{k+2} - 2kc_k + c_k)x^k$$

B $$\sum_{k=1}^{\infty} ((k+2)c_k - 2kc_k + c_k)x^k$$

C $$c_0 + 2c_2 + \sum_{k=1}^{\infty} ((k+2)(k+1)c_{k+2} - 2kc_k + c_k)x^k$$

D $$c_0 + 2c_2 + \sum_{k=1}^{\infty} ((k+2)c_k - 2kc_k + c_k)x^k$$

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Solution: C

The first and third series begin with the x^0 term, but the second series begins with the x^1 term, so we'll pull out the x^0 term from both the first and the third series.

$$2(2-1)c_2x^{2-2} + \sum_{n=3}^{\infty} n(n-1)c_nx^{n-2} - 2\sum_{n=1}^{\infty} nc_nx^n + c_0x^0 + \sum_{n=1}^{\infty} c_nx^n$$

$$c_0 + 2c_2 + \sum_{n=3}^{\infty} n(n-1)c_nx^{n-2} - 2\sum_{n=1}^{\infty} nc_nx^n + \sum_{n=1}^{\infty} c_nx^n$$

Each series now begins with the x^1 term, so they're in phase. To make their indices match, we'll make the substitution $k = n - 2$ and $n = k + 2$ into the first series, and make the substitution $k = n$ into the second and third series.

$$c_0 + 2c_2 + \sum_{k=1}^{\infty} (k+2)(k+2-1)c_{k+2}x^k - 2\sum_{k=1}^{\infty} kc_kx^k + \sum_{k=1}^{\infty} c_kx^k$$

$$c_0 + 2c_2 + \sum_{k=1}^{\infty} (k+2)(k+1)c_{k+2}x^k - 2\sum_{k=1}^{\infty} kc_kx^k + \sum_{k=1}^{\infty} c_kx^k$$

The series are now in phase and the indices match, so we'll add them and then factor out x^k.

$$c_0 + 2c_2 + \sum_{k=1}^{\infty} (k+2)(k+1)c_{k+2}x^k - 2kc_kx^k + c_kx^k$$

$$c_0 + 2c_2 + \sum_{k=1}^{\infty} ((k+2)(k+1)c_{k+2} - 2kc_k + c_k)x^k$$

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